Solar Installation Management System

Build the digital backbone of a solar company that has outgrown WhatsApp quotes and site registers.

1) The Challenge

Your task is to build **one unified software platform** that solves the real, day-to-day operational headaches of a growing rooftop solar company. Think of it as the **digital backbone** that takes a customer **from the first quote request to the 20-year warranty**. Read the scenes that follow as things that actually happen on an ordinary week, and design the system that makes each of them easy.

2) Meet SunPower Solutions

SunPower is a fast-growing solar installation company that completes **about 40 rooftop projects a month** for homes and small factories. It runs **roof surveys**, keeps **high-value stock** such as panels, inverters and batteries, and sends **a crew of 15 technicians** to customer sites. After installation, **an endless stream of service calls** follows. It sells ready-made systems by capacity, **from 1 kW to 5 kW**, in two kinds: **On-Grid** (connected to the electricity grid) and **Off-Grid** (with battery backup).

Packages

- **Solar Panel Set (Hardware):** Solar panels, connectors and wires

- **On-Grid System (1 kW to 5 kW):** Inverter (turns sun power into home electricity), wiring boxes, battery charge controller, panels, safety earthing and metal stands
- **Off-Grid System (1 kW to 5 kW):** Everything in On-Grid, plus a large battery that stores power
- **Installation (Service):** Fitting and testing by our crew at the customer's site

How Things Run Today

- **Customer enquiries:** Arrive on WhatsApp, calls and a web form, and get lost
- **Quotes:** Typed by hand each time, with parts often missed
- **Serial numbers:** Noted on paper, and lost by the time a warranty claim comes
- **Site work:** Recorded in site registers, with photos scattered on phones
- **Revenue and margins:** No clear picture anywhere

The owner is losing money, customers are frustrated, and technicians are overwhelmed.

3) A Week at SunPower

The scenes below follow one week at SunPower, from a homeowner asking for a quote to the director reviewing the month. **Every one of them is a problem someone at the company faces today.**

The rooftop inquiry

A homeowner fills in a form online asking for a 3 kW system under the subsidy. Their name, phone, address and needs should **land with the sales team straight away**, and **the enquiry should never vanish**.

Someone must call them and check the basics: **how much shadow-free roof space they have, what their monthly electricity bill is, and whether the grid connection is suitable**. If it looks good, a technician visits to measure the **roof direction, tilt and cable distance**. **Only after the survey is approved should the customer receive a quote.**

Getting the kit right

The customer says yes to a 3 kW system. The **quote should come from a ready template**, and the **exact list of parts should follow on its own**: panels, inverter, stands, switches, wires and safety parts. **Nobody should have to remember every small part.**

A customer usually **pays an advance, or the full amount**, before work starts. Once the order is confirmed, **the warehouse should know what to pack** and **the installation crew should see the job on their schedule**, without anyone phoning to tell them.

Dispatch day and serial numbers

The truck is about to leave with ₹1,50,000 worth of inverters and batteries. Every inverter, battery and charge controller **has its own serial number**, and each **must be scanned before it leaves the warehouse** so the company knows exactly **which unit went to which customer**.

The moment a delivery is done, **the warranty clock should start by itself** for every serial number: for example **2 years for an inverter or battery, and 20 years for solar panels**. No one should have to type dates into a register.

On the roof

The crew mounts the stands, lays the panels, connects the boxes and finishes the earthing. They test that power is being generated, then open a **simple form on their phone or tablet**.

There they note the **start and finish times, confirm the system size, and upload photos** of the finished roof. The homeowner **signs on the screen to accept the work**. The company should have **clear proof of the job**, and the customer should have no doubt about what was installed.

The rainy season breakdown

Eighteen months later, it is pouring and a customer calls: the inverter is showing a fault code and there is zero power. Support should **find the customer and the exact inverter serial number within seconds**, and see whether it is **still under warranty**.

If it is, the repair or replacement should **cost the customer nothing**. If not, the parts and labour should be **billed automatically**. Either way the complaint should be **tracked as a ticket until the customer confirms that everything works**.

The director's month-end

The director sits down and asks a simple question: **which installations made money, which lost money, and what is still unfinished?** Advance payments, stock purchases, technician hours and unfinished steps for customers are **scattered across notebooks and phones**.

There are also **invoices to send, government subsidy paperwork to track, and technicians to pay**. The director wants to see how the company is doing **today, this week and this month**, including **the profit on each job and the repairs still open**, and to **share these numbers** when needed.